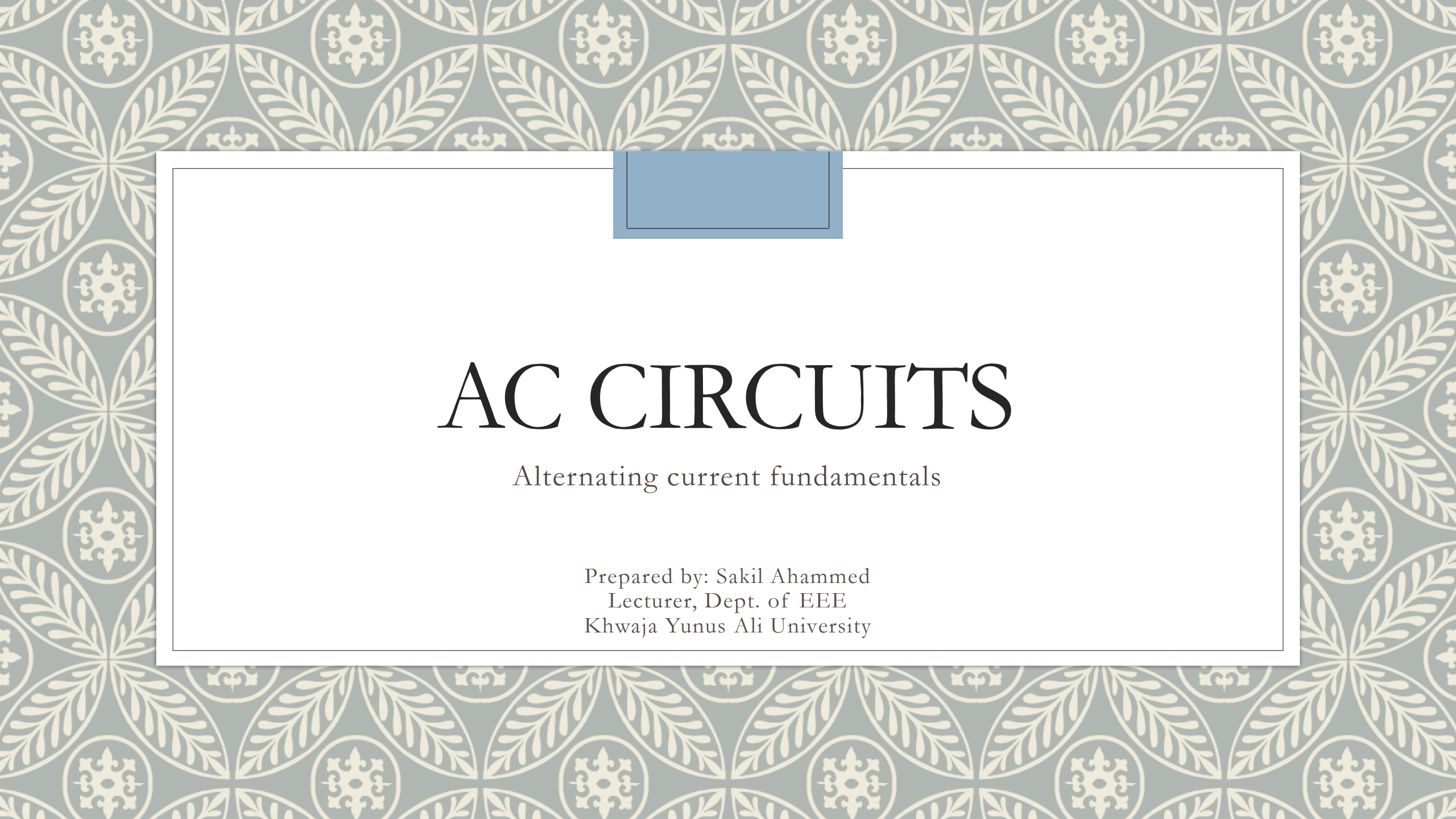




AC CIRCUITS

Alternating current fundamentals

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Brief History

- At the end of 19th century, which was the better – ac or dc – was hotly debated and had extremely outspoken supporters on both sides.
- The dc side was led by Thomas Edison, who had earned a lot of respect due to his contributions.
- Power generation using AC starts from the contribution of Tesla. The commercial success in ac came from George Westinghouse and his team, including Tesla, C.F. Scott, B.G. Lamme and more.
- Most significant contribution to AC was the patenting of the polyphase ac motor by Tesla in 1888. The induction Motor and polyphase generation and distribution system doomed the used of AC as primary energy.
- Till 1800s, DC was really popular. But, Ac is efficient and economical to transmit over long distances, thus ac ended up the winner.



Nikola Tesla

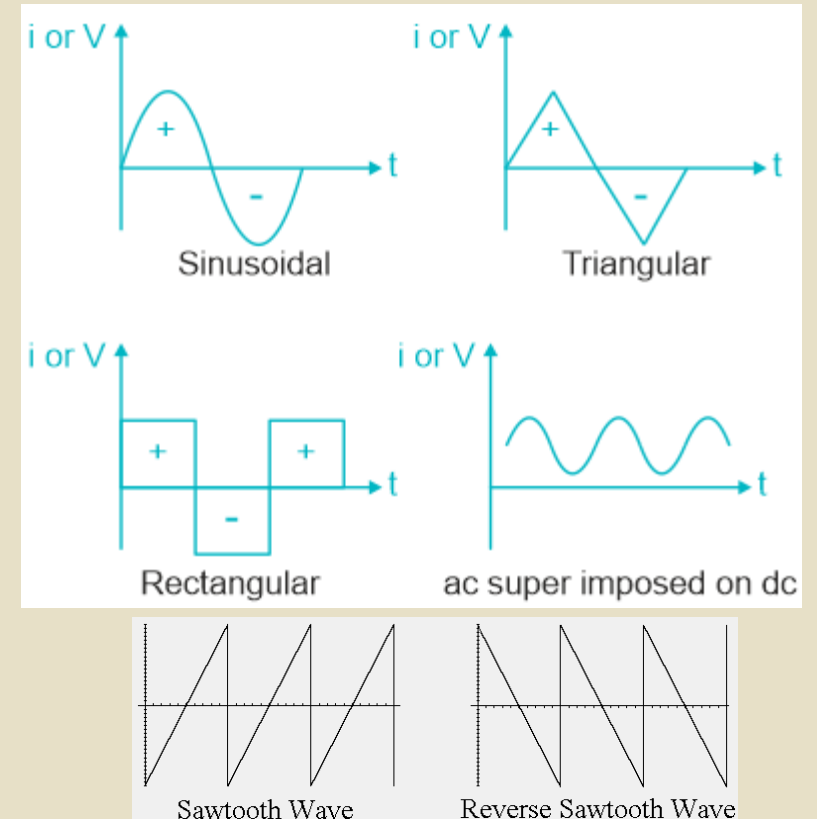
AC Circuit: Definition and Types

- The quantity that changes continuously in magnitude between zero and a maximum value with alternating directions at regular time intervals is known as an alternating quantity such as current (I) or voltage (V).
- AC circuits are powered by an alternating source such as alternating current or voltages which are sinusoidal and change periodically in direction and magnitude.
- (AC quantity- Variation of magnitude with respect to time)

Types of AC signals:

1. Sinusoidal Signals (Sine and Cosine wave)
2. Triangular wave signals
3. Sawtooth wave signals
4. Square wave signals

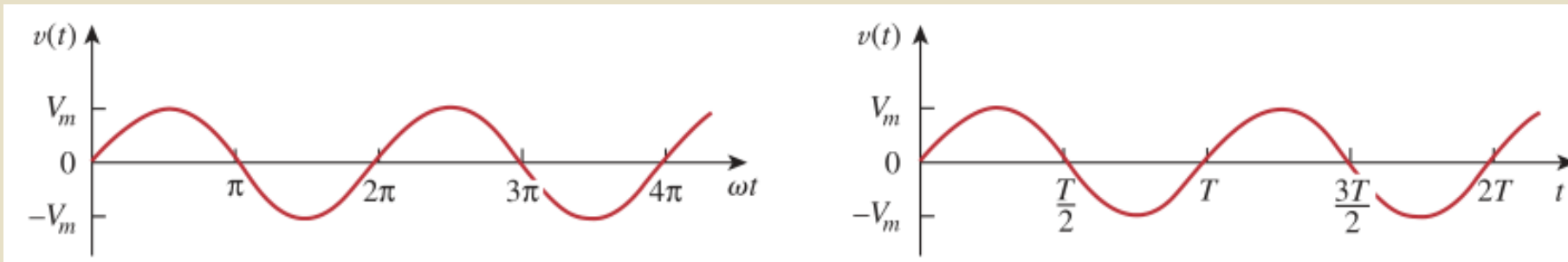
And many more.



Why sinusoids among all signals?

Among all of the signals, we mainly will use sinusoidal signals. Because-

1. Nature itself is characteristically sinusoidal. (Motion of pendulum, vibration of a string, ripples on the ocean surface, natural response of under damped second-order systems, etc.)
2. Sinusoidal signals are easy to generate and transmit. In both communication and electric power industry, it is mainly used.
3. Any practical periodic signals can be represented by a sum of sinusoids. (To know more about Fourier series representation of non-sinusoidal waveform, visit [here](#))
4. Sinusoids are easy to handle mathematically. Derivative and Integration of sinusoids are itself sinusoids.



Some basic definitions

Periodic Function: A function returning to the same value at regular intervals. satisfies $f(t)=f(t+nT)$, for all t and for all integers n .
Example: Sinusoidal, triangular, sawtooth, square signals and many more.

Time period: The time taken for one complete oscillation to occur is called the Time Period. It is denoted by T . Its unit is seconds.

Frequency: Frequency is the number of vibrations or the number of occurrences of a repeating event per unit of time. It is denoted by f . It's unit is Hz or s^{-1} .

$$\text{Mathematically, } f = \frac{1}{T}$$

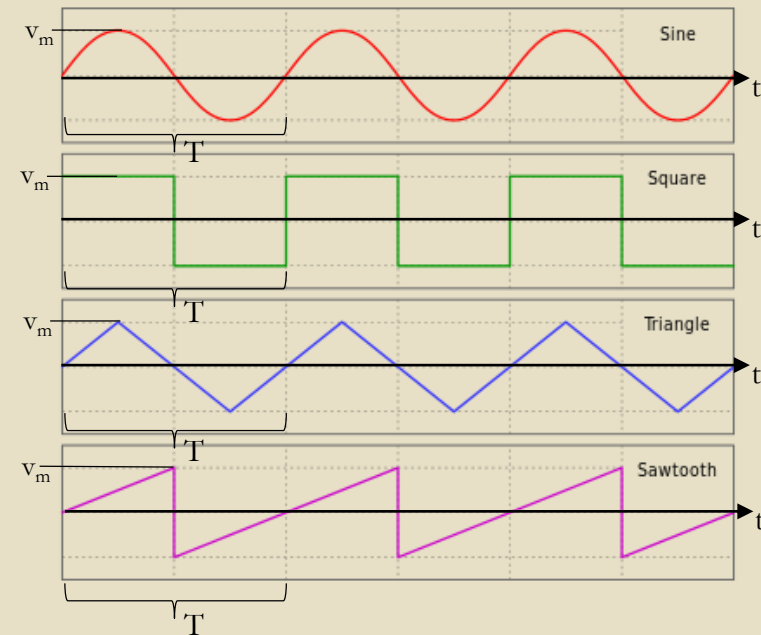
Amplitude/Magnitude: Magnitude refers to the size or amount of a physical quantity, while amplitude refers to the strength or intensity of a wave. V_m is amplitude of the signals.

Angular Frequency(ω): The angular frequency refers to the angular displacement of any wave element per unit of time or the rate of change of the waveform phase.

Phase angle: Φ (phi) – is the phase angle in degrees or radians that the waveform has shifted either left or right from the reference point.

Phase difference: the difference in phase between two sinusoidally varying quantities that have the same frequency, measured either as an angle or a time.

Unit, rad/sec. Mathematically, $\omega = 2\pi f = \frac{2\pi}{T}$



Periodicity and Phase relations

Consider the sinusoidal voltage, $v(t) = V_m \sin \omega t$

Where, V_m = the amplitude of the sinusoid

ω = the angular frequency in radians/s

ωt = the argument of the sinusoid

From the waveform, T = Time period of the signal.

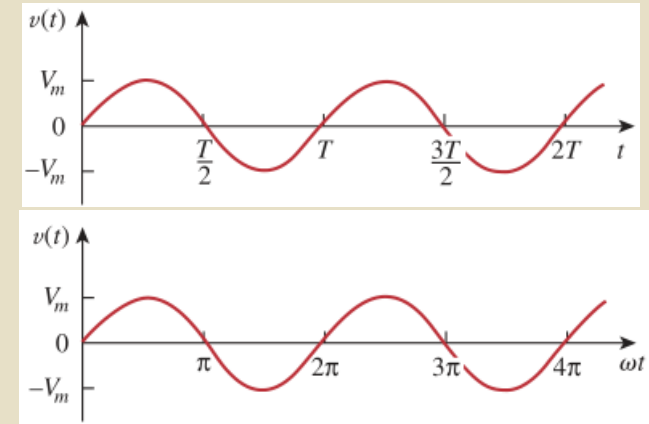
$$v(t + T) = V_m \sin \omega(t + T) = V_m \sin \omega \left(t + \frac{2\pi}{\omega} \right) = V_m \sin(\omega t + 2\pi) = V_m \sin \omega t = v(t)$$

Thus, $v(t + T) = v(t)$, and the function is periodic.

If phase angle exists, we can write the equation of voltage,

$$v(t) = V_m \sin(\omega t + \Phi)$$

Where, Φ = phase angle expressed in radian or degree.



Phase difference

Let us examine two sinusoids,

$$v_1(t) = V_m \sin(\omega t + 0) = V_m \sin \omega t \quad \text{and} \quad v_2(t) = V_m \sin(\omega t + \Phi)$$

Therefore, v_2 leads v_1 by $(\Phi - 0) = \Phi$. Or v_1 lags v_2 by Φ .

Here, v_1 and v_2 are out of phase.

If $\Phi = 0$, then v_1 and v_2 are in phase as they reach their minima and maxima at the same time.

- We can compare v_1 and v_2 only when they operate at the same frequency although they are in different magnitude.
- Sinusoids can be converted from sine form to cosine form and vice versa.

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

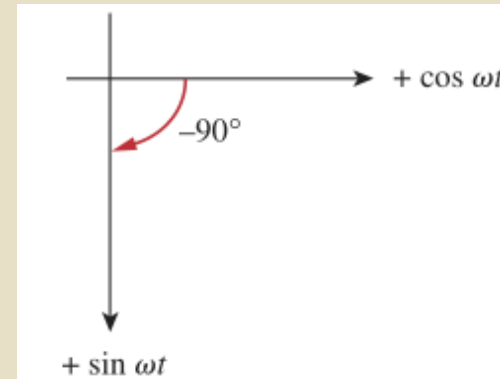
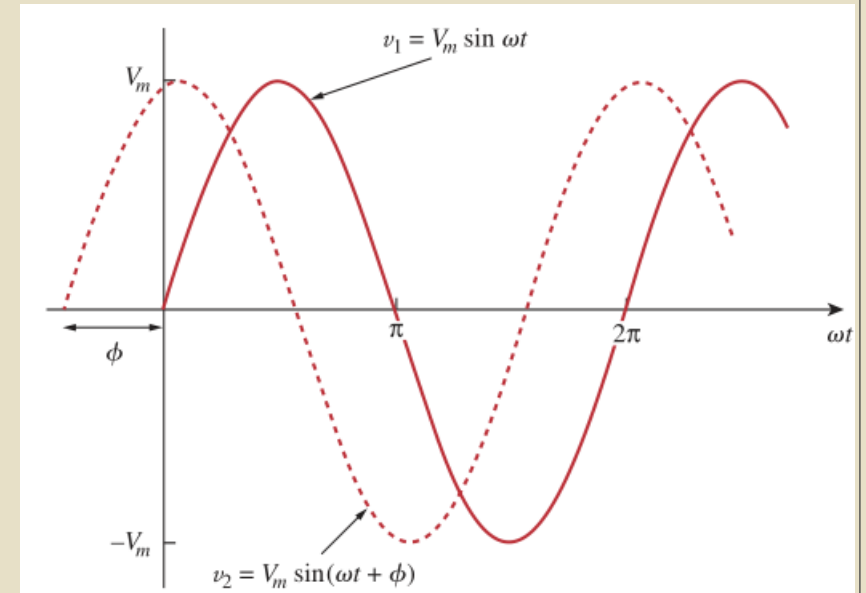
$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\sin(\omega t \pm 180^\circ) = -\sin \omega t$$

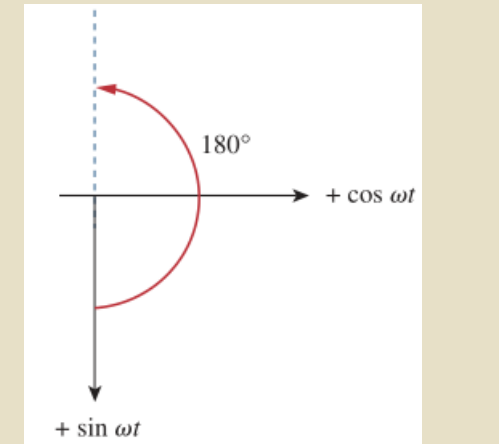
$$\cos(\omega t \pm 180^\circ) = -\cos \omega t$$

$$\sin(\omega t \pm 90^\circ) = \pm \cos \omega t$$

$$\cos(\omega t \pm 90^\circ) = \mp \sin \omega t$$



$$\cos(\omega t - 90^\circ) = \sin \omega t,$$



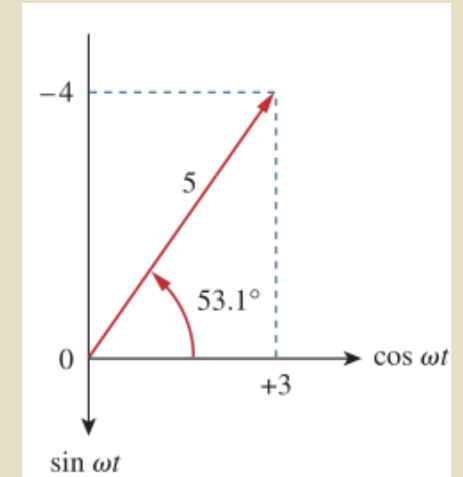
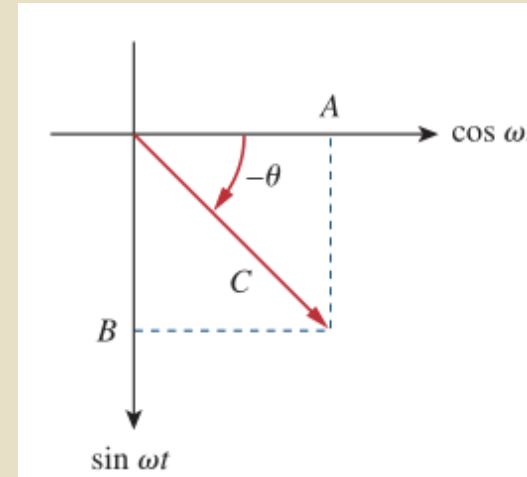
$$\sin(\omega t + 180^\circ) = -\sin \omega t.$$

Summation of Signals

- If $v_1(t) = A \sin \omega t$ and $v_2(t) = B \cos \omega t$
- Then, $v = v_1 + v_2 = A \sin \omega t + B \cos \omega t = C \cos(\omega t - \theta)$
- Where, $C = \sqrt{A^2 + B^2}$, and $\theta = \tan^{-1} \frac{B}{A}$

For example,

$$3 \cos \omega t - 4 \sin \omega t = 5 \cos(\omega t + 53.1^\circ)$$



Practice Problems:

1. Find the amplitude, phase angle, period, and frequency of the sinusoid $v(t) = 12 \cos(50t + 10^\circ)$

Ans: 12V, 10° , 0.1257 sec, 7.958 sec.

2. Given the sinusoid $30 \sin(4\pi t - 75^\circ)$ calculate its amplitude, phase, angular frequency, period, and frequency.

Answer: 30, -75° , 12.57 rad/s, 0.5 s, 2 Hz.

3. Calculate the phase angle between $v_1 = -10 \cos(\omega t + 50^\circ)$ and $v_2 = 12 \sin(\omega t - 10^\circ)$ State which sinusoid is leading.

Ans: v_2 leads v_1 by 30° .

4. Find the phase angle between

$$i_1 = -4 \sin(377t + 55^\circ) \quad \text{and} \quad i_2 = 5 \cos(377t - 65^\circ)$$

Does i_1 lead or lag i_2 ?

Answer: 210° , i_1 leads i_2 .

Phasors

- Sinusoids are easily expressed in terms of phasors, which is more convenient to work with.
- **A phasor is a complex number that represents the amplitude and phase of a sinusoid.**
- The notion of solving ac circuits using phasors was first introduced by Charles Steinmetz in 1893.
- Complex Number: In Rectangular form, $z = x + jy$ where, $j = \sqrt{-1}$
- Here, x is real part and y is imaginary part of complex plane. It is similar to two axis of two dimensional vectors.
- In polar form, $z = r\angle\phi = re^{j\phi}$ (exponential form)
- Where, r= magnitude of z, and ϕ is the phase of z.
- $r = \sqrt{x^2 + y^2}$, $\phi = \tan^{-1} \frac{y}{x}$
- On the other hand, $x = r \cos \phi$, $y = r \sin \phi$
- Thus, $z = x + jy = r\angle\phi = r(\cos \phi + j \sin \phi)$

Phasor Operations

Given complex numbers, $z = x + jy = r\angle\phi$, $z_1 = x_1 + jy_1 = r_1\angle\phi_1$ and $z_2 = x_2 + jy_2 = r_2\angle\phi_2$

Addition: $z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2)$

Subtraction: $z_1 - z_2 = (x_1 - x_2) + j(y_1 - y_2)$

Multiplication: $z_1 z_2 = r_1 r_2 \angle \phi_1 + \phi_2$

Division: $\frac{z_1}{z_2} = \frac{r_1}{r_2} \angle \phi_1 - \phi_2$

Reciprocal: $\frac{1}{z} = \frac{1}{r} \angle -\phi$

Square Root: $\sqrt{z} = \sqrt{r} \angle \phi/2$

Complex Conjugate: $z^* = x - jy = r \angle -\phi = r e^{-j\phi}$

Note: $\frac{1}{j} = -j$

Euler's identity: $e^{\pm j\phi} = \cos\phi \pm j\sin\phi$

We may write, $\cos\phi = \text{Re}(e^{j\phi})$

$\sin\phi = \text{Im}(e^{j\phi})$

Re- real part, Im- imaginary part.

Given a sinusoidal function, $v(t) = V_m \cos(\omega t + \phi)$.

$v(t) = V_m \cos(\omega t + \phi) = \text{Re}(V_m e^{j(\omega t + \phi)})$

Or,

$v(t) = \text{Re}(V_m e^{j\phi} e^{j\omega t})$

Thus,

$v(t) = \text{Re}(\mathbf{V} e^{j\omega t})$

Where,

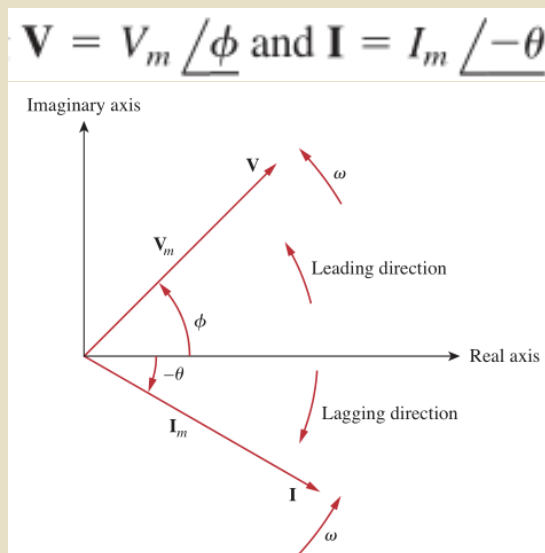
$\mathbf{V} = V_m e^{j\phi} = V_m \angle \phi$

Phasor representation

- Neglecting time factor $e^{j\omega t}$, we get

$$\begin{array}{ccc} v(t) = V_m \cos(\omega t + \phi) & \Leftrightarrow & \mathbf{V} = V_m \angle \phi \\ \text{(Time-domain representation)} & & \text{(Phasor-domain representation)} \end{array}$$

- If, $\mathbf{V} = V_m \angle \phi$ and $\mathbf{I} = I_m \angle -\theta$



Time domain representation	Phasor domain representation
$V_m \cos(\omega t + \phi)$	$V_m \angle \phi$
$V_m \sin(\omega t + \phi)$	$V_m \angle \phi - 90^\circ$
$I_m \cos(\omega t + \theta)$	$I_m \angle \theta$
$I_m \sin(\omega t + \theta)$	$I_m \angle \theta - 90^\circ$

Difference between $v(t)$ and V

1. $v(t)$ is the instantaneous or time domain representation, while V is the frequency or phasor domain representation.
2. $v(t)$ is time dependent, while V is not.
3. $v(t)$ is always real with no complex term, while V is generally complex.

Practice problem

1. Evaluate the complex numbers:
- (a) $(40\angle 50^\circ + 20\angle -30^\circ)^{1/2}$ Ans: a) $6.91\angle 12.81^\circ$
 (b) $\frac{10\angle -30^\circ + (3 - j4)}{(2 + j4)(3 - j5)^*}$ b) $0.565\angle -160.13^\circ$

4. Express these sinusoids as phasors:
 (a) $v = 7 \cos(2t + 40^\circ)$ V
 (b) $i = -4 \sin(10t + 10^\circ)$ A
- Answer:** (a) $\mathbf{V} = 7\angle 40^\circ$ V, (b) $\mathbf{I} = 4\angle 100^\circ$ A.

2. Evaluate the following complex numbers:
 (a) $[(5 + j2)(-1 + j4) - 5\angle 60^\circ]^*$
 (b) $\frac{10 + j5 + 3\angle 40^\circ}{-3 + j4} + 10\angle 30^\circ + j5$
- Answer:** (a) $-15.5 - j13.67$, (b) $8.293 + j7.2$

5. Find the sinusoids corresponding to these phasors:

- (a) $\mathbf{V} = -25\angle 40^\circ$ V
 (b) $\mathbf{I} = j(12 - j5)$ A

Answer: (a) $v(t) = 25 \cos(\omega t - 140^\circ)$ V or $25 \cos(\omega t + 220^\circ)$ V,
 (b) $i(t) = 13 \cos(\omega t + 67.38^\circ)$ A.

3. Transform these sinusoids to phasors:

- (a) $i = 6 \cos(50t - 40^\circ)$ A
 (b) $v = -4 \sin(30t + 50^\circ)$ V

Ans: $\mathbf{I} = 6\angle -40^\circ$ A $\mathbf{V} = 4\angle 140^\circ$ V

6. Given $i_1(t) = 4 \cos(\omega t + 30^\circ)$ A and $i_2(t) = 5 \sin(\omega t - 20^\circ)$ A, find their sum.

Ans: $i(t) = 3.218 \cos(\omega t - 56.97^\circ)$ A

7. If $v_1 = -10 \sin(\omega t - 30^\circ)$ V and $v_2 = 20 \cos(\omega t + 45^\circ)$ V, find $v = v_1 + v_2$.

Answer: $v(t) = 29.77 \cos(\omega t + 49.98^\circ)$ V.

8. Find the voltage $v(t)$ in a circuit described by the integrodifferential equation

$$2 \frac{dv}{dt} + 5v + 10 \int v dt = 50 \cos(5t - 30^\circ)$$

using the phasor approach.

Answer: $v(t) = 5.3 \cos(5t - 88^\circ)$ V.

RMS and Average value Calculation

- The Average value of a signal $x(t)$ can be written as,

$$x_{avg}(t) = \frac{1}{T} \int_0^T x(t) dt$$

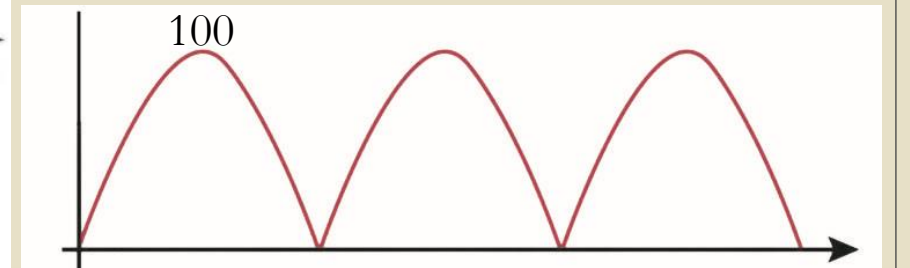
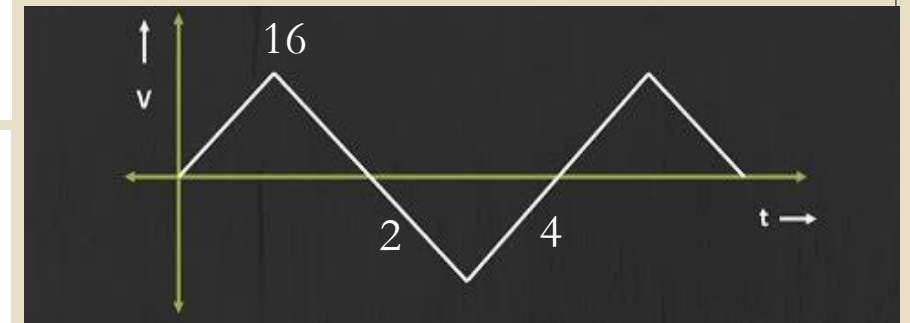
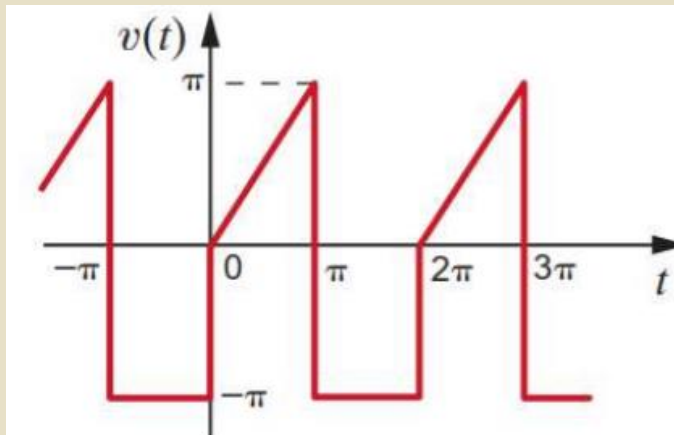
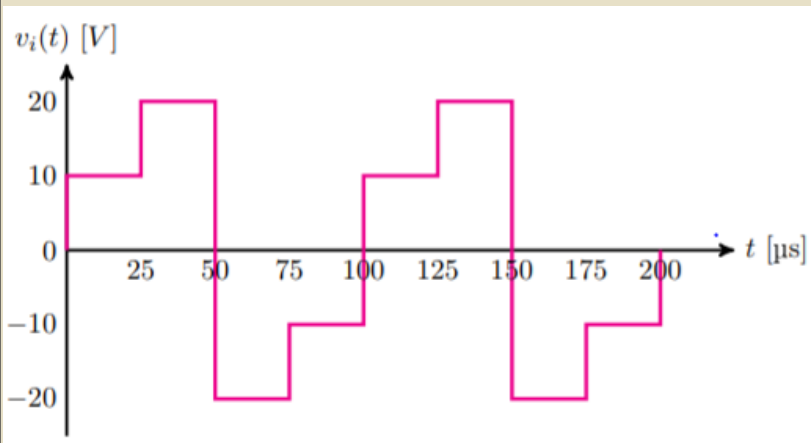
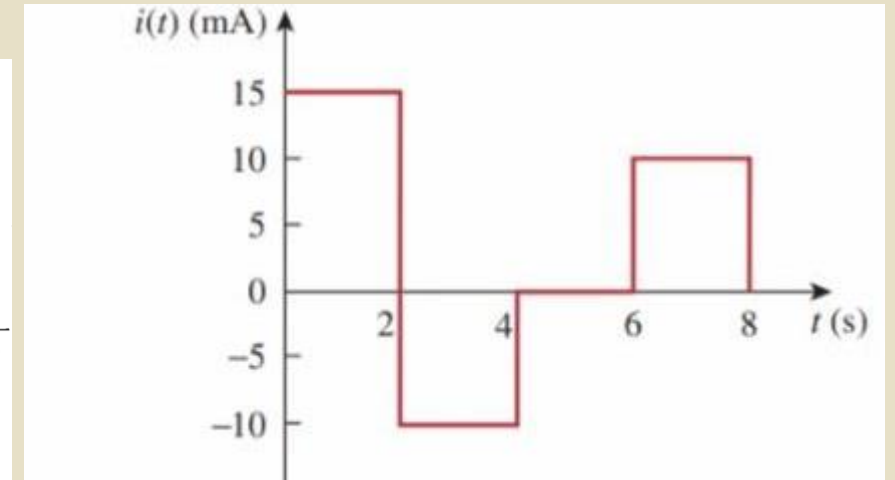
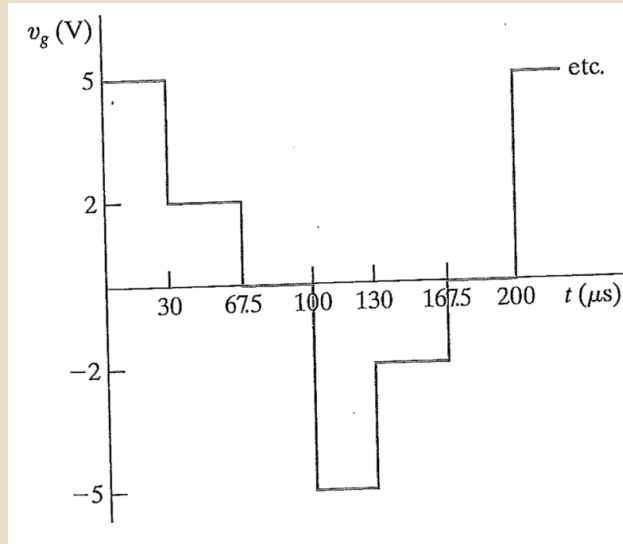
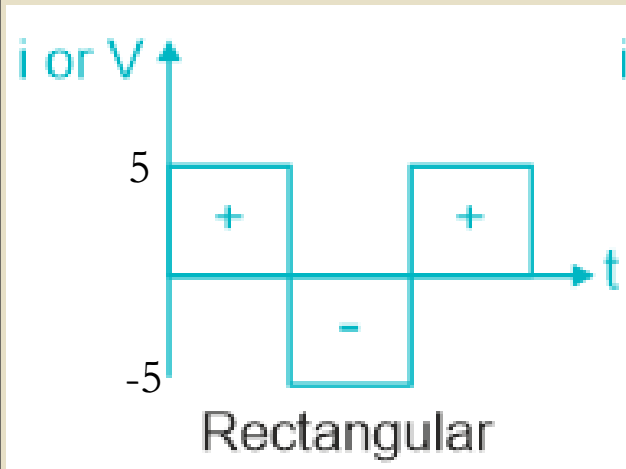
Where, T= Time period of the signal.

- The RMS value of signal $x(t)$ can be written as,

$$x_{rms}^2 = \frac{1}{T} \int_0^T x^2(t) dt$$

Practice Problem

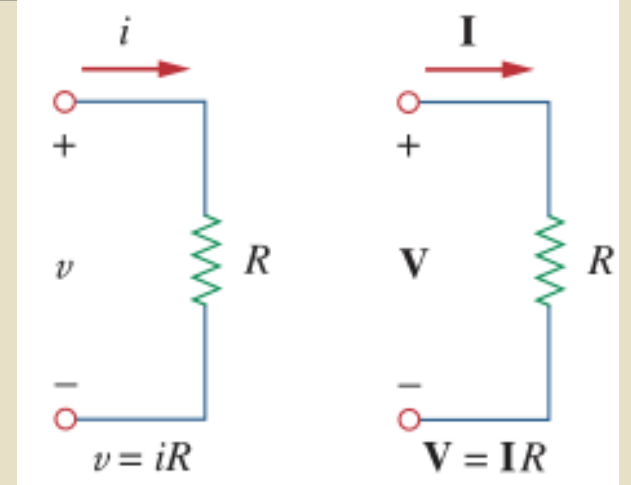
1. Calculate RMS and average value of the following figures,
(For both half cycle and full cycle)



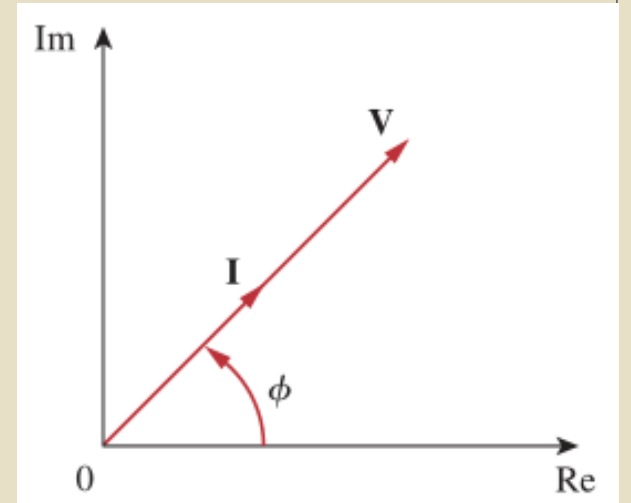
AC through Resistance

If $i = I_m \cos(\omega t + \phi)$ passes through resistor R , then
Voltage $v = iR = RI_m \cos(\omega t + \phi) = RI_m \angle \phi = RI$

Element	Time domain	Frequency domain
R	$v = Ri$	$V = RI$



Time and frequency domain relation of voltage and current



Phasor diagram for resistor

Next: Inductance and Capacitance